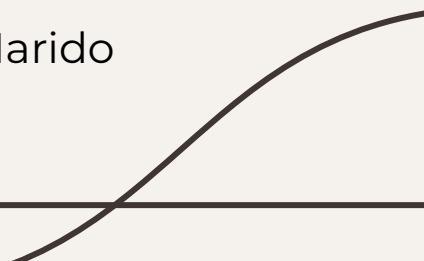


Predictive Maintenance using Machine Learning

By Team B: Abigail Duran, Brandon Rodriguez, Miguel Narido



Why Predictive Maintenance?

- Unexpected machine failures cause downtime and increased maintenance costs.
- Predictive maintenance uses operational data to identify potential equipment failures.
- Machine learning can identify patterns associated with equipment failure.
- Goal: Develop a model to classify machine operating conditions as failure or no failure.

Our Dataset

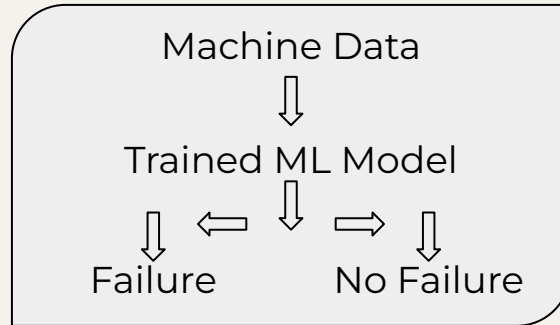
- Source: [UCI AI4I Predictive Maintenance Dataset](#)
- Samples: 10,000
- Features: 6 input variables
- Missing values: 0
- Main Target Variable: Machine failure
- Data Type: Synthetic industrial machine data
- Target Categories: Failure (1) / No Failure (0)

Dataset Features

- Product Type: Low (L), Medium (M), or High quality (H).
- Air Temperature: Kelvin (K)
- Process Temperature: Kelvin (K)
- Rotational Speed: Revolutions Per Minute (RPM)
- Torque: Newton-meters (Nm)
- Tool Wear: Minutes (min)

How Will We Predict Machine Failure?

- Supervised machine learning
- Binary classification: Failure or No Failure
- Potential Models:
 - Logistic Regression
 - Decision Tree
 - Random Forest
- Evaluate using Precision, Recall and F1-score.



Previous Work

- Matzka (2020) - Original Dataset Study
 - Introduced the AI4I 2020 Predictive Maintenance Dataset.
 - Used Decision Trees and explainable AI to predict and analyze machine failures.
 - Demonstrated how machine operating conditions can help identify potential equipment failures.
- Gonzalez Islas et al. (2026) - ML Model Comparison
 - Compared multiple algorithms, including Random Forest, Decision Tree, and Artificial Neural Networks (ANN).
 - Investigated feature selection and data balancing techniques to improve failure classification.
 - Evaluated models using accuracy, precision, recall, and F1-score.
- How This Relates to Our Project
 - We plan to compare different ML models to predict machine failure.
 - We will evaluate how accurately each model identifies failures.

Our Project Plan

1. Explore and preprocess the dataset
2. Select relevant features
3. Split data into training and testing sets
4. Train classification models
5. Evaluate and compare model performance
6. Identify which operating features are most associated with machine failure.